RV College of Engineering CIE - I

DEPARTMENT OF MATHEMATICS

Academic year 2024-2025 (Odd Semester 2024)

Date	21.10.2024	Time	9.30 AM– 11.30 AM
TEST	Quiz- I & Test -I	Maximum Marks	10+50
Course Title	STATISTICS, LAPLACE TRANSFORM AND NUMERICAL METHODS	Course Code	MA231TB
Semester	III	Programs	AS, BT, CH, ME, IEM

Sl. No	Instructions: Answer all questions. Part - A	M	CO	BT
1	For a statistical data with $n=5, \sum x=18, \sum y=15, \sum x^2=110$ and $\sum xy=71$, the best fit straight line by the method of least squares is ________.	1	1	1
2	For a distribution the mean is 12 and the variance is 19. Then the second moment about the mean is	1	2	1
3	The first two moments about the point $x=2$ are 1 and 16, then the second central moment is ________.	2	2	2
4	The equations of regression lines are $y=0.5x+p$ and $y=0.4y+q$, then the correlation coefficient is ________.	2	2	2
5	$L[\sin 3t+\cosh 5t]=$ ________.	1	1	1
6	$L[(t^2e^{3t})]=$ ________.	1	2	1
7	$L[(e^{-5t}+e^{t})^2]=$ ________.	2	1	2
	Part - B			
1	Determine the equation of the regression plane connecting x_1, x_2 and y, given (table below). Estimate y at $x_1=1.8$ and $x_2=112$.	10	4	3
2	The marks obtained by 50 students in a Mathematics test are given below (table below). Compute first four moments about the mean and also find the measures β_1 and β_2 for the following distribution and comment on the nature of distribution of students.	10	3	
3a	The following pairs of observations were noted in an experimental work on cosmic rays. Find by the method of least squares the most possible values of a and b for the equation $y=a+bx$ which fits the data (table below). Predict y when $x=3.75$.	6	1	

Part B, Q1:

Diffusion time (hours) x_1	1.5	2.5	0.5	1.2	2.6	0.3
Sheet – resistance (Ohms – cm) x_2	66	87	69	141	93	105
Current gain (y)	5.3	7.8	7.4	9.3	10.8	9.1

Part B, Q2:

Marks	10-20	20-30	30-40	40-50	50-60	60-70
No. of students	4	8	10	15	8	5

Part B, Q3a:

x	1	2	3	4	5	6
y	2.98	4.26	5.21	6.10	6.80	7.50

Academic year 2024-2025 (Odd Semester 2024)

3b	The Coefficient of rank correlation obtained by ten students in statistics and accountancy was 0.2 . It was later discovered that the difference in ranks in the two subjects of one of the students was wrongly taken as 9 instead of 7. Find the correct rank correlation coefficient.	4	2
4	A study was made on the amount of converted sugar in a certain process at various temperatures. The data were coded and recorded as follows:	10	3

x	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0
y	7.8	8.5	9.8	9.5	8.9	8.6	10.2	9.3	9.2	10.5

Construct the table of values and compute the correlation coefficient for this data and also find the regression lines of x on y and y on x.

5	Determine the transformation in frequency domain if the signal in time domain is given by $f(t) = \frac{cos6t - cos3t}{t} + e^{-5t} sin2t\, cos3t + t(\sqrt{t} + 1)^2$.	10	2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test Max Marks	6	14	20	10	6	14	30			

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Date	
Course Code	Materia
Course Name	

Q. No.	
1	How many electrons can occu
2	The principal quantum numb
3	In which type of bond do at
4	A large energy gap between type of materials
5	What is the basic repeating
6	Single atom or ion missing defect in a crystal structur
7	The fractional increase in
8	Identify the thermoelec different materials or jun
9	Define dielectric strengt

1	a) Describe Pauli Excl

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DEPARTMENT OF CIVIL ENGINEERING

Date	8^{th} Jan 2024		
Course Code	CV232AT	Maximum Marks	50
Sem	III Semester	Duration (minutes)	90
		CIE – I	
	Environment and Sustainability		

No.	Questions	M	BT	CO
1	a) Define the terms i) Environment ii) Ecology iii) Bio-Diversity	6	1	1
	b) Materials flow in eco system is cyclic and Energy flow is unidirectional, justify.	4	2	1
2.	Classify different types of ecosystems. Explain in brief any two ecosystems with their structure and composition	10	2	2
3.	a) Recall the concept of ecological succession and classify the same	5	1	1
	b) Discuss the approaches of conserving the biodiversity	5	2	1
4.	a) Highlight the salient features of various hotspots of biodiversity in India	5	1	1
	b) Briefly discuss the causes, effects and preventive measures of water pollution	5	2	2
5.	Briefly describe the various Environmental protection acts in India	10	2	2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	25	25	--	-	16	34	-	-	-	-

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DEPARTMENT OF CIVIL ENGINEERING

Date	8th Jan 2024	Maximum Marks	50
Course Code	CV232AT	Duration (minutes)	90
Sem	III Semester	CIE – I	
Environment and Sustainability			

o.	Questions	M	BT	CO
1	a) Define the terms i) Environment ii) Ecology iii) Bio-Diversity	6	1	1
	b) Materials flow in eco system is cyclic and Energy flow is unidirectional, justify.	4	2	1
2.	Classify different types of ecosystems. Explain in brief any two ecosystems with their structure and composition	10	2	2
3.	a) Recall the concept of ecological succession and classify the same	5	1	1
	b) Discuss the approaches of conserving the biodiversity	5	2	1
4.	a) Highlight the salient features of various hotspots of biodiversity in India	5	1	1
	b) Briefly discuss the causes, effects and preventive measures of water pollution	5	2	2
5.	Briefly describe the various Environmental protection acts in India	10	2	2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	25	25	--	-	16	34	-	-	-	-

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3b	The Coefficient of rank correlation obtained by ten students in statistics and accountancy was 0.2 . It was later discovered that the difference in ranks in the two subjects of one of the students was wrongly taken as 9 instead of 7. Find the correct rank correlation coefficient.	4	2
4	A study was made on the amount of converted sugar in a certain process at various temperatures. The data were coded and recorded as follows:	10	3
5	Determine the transformation in frequency domain if the signal in time domain is given by $f(t) = \frac{cos6t-cos3t}{t} + e^{-5t} sin2t\,cos3t + t(\sqrt{t}+1)^2$.	10	2

x	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0
y	7.8	8.5	9.8	9.5	8.9	8.6	10.2	9.3	9.2	10.5

Construct the table of values and compute the correlation coefficient for this data and also find the regression lines of x on y and y on x.

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks	Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
Distribution	Test Max Marks	6	14	20	10	6	14	30			

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	2
Date	
Course Code	
Course Name	Materia

Q. No.	
1	How many electrons can occ
2	The principal quantum numb
3	In which type of bond do at
4	A large energy gap between type of materials
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1	a) Describe Pauli Excl

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DEPARTMENT OF
BIOTECHNOLOGY
Academic year 2023-2024 (Odd Sem)

...e	January 8, 2024	Maximum Marks	50
...rse Code	BT232AT	Duration	90 min
...a	III Semester	Closed Book Offline Test-1	

Bio Safety Standards and Ethics (Basket course)

...o	Questions (Test)	M	BT-L	CO
1.	Elaborate on various bio hazards from nature	10	2	1
2	List out and discuss biosafety levels with a neat triangular diagram	10	4	1
3	Explain with a help of a neat diagram, biosafety cabinet B2	10	4	1
4	Differentiate the bio safety cabinets I, II and III	10	2	1
5	Give details on GMO and their applications in various fields	10	2	2

...T-L-Blooms Taxonomy, CO-Course Outcomes, M-Marks

...rks ...tribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	40	10	---	---	---	---	30	20	---	...

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Department of Civil Engineering

Date	21/10/2024	Maximum Marks	50
Course Code	CV232AT	Duration	90 Min +20 Min
Sem	III	CIE – I (Quiz + Test)	
Environment and Sustainability (Basket Course)			

Part A

Sl. No.	Questions	M	BT	CO
1.	Distinguish between ecology and ecosystem.	2	1	1
2.	What is primary and secondary succession?	2	1	1
3.	Write any four methods of disposal of solid waste.	2	1	2
4.	What is the importance of safety in workplace?	2	1	2
5.	List any four environmental protection acts in India.	2	1	1

Part B

Sl. No.	Questions	M	BT	CO
1.	a. Briefly explain the structure and components of a ecosystem.	5	2	1
	b. Materials flow in eco system is bi directional and Energy flow is unidirectional justify.	5	2	1
2.	With a neat sketch explain the nitrogen cycle.	10	2	1
3.	Define solid waste management and list the objectives of solid waste management.	10	2	2
4.	Define Air Pollution? Classify and explain the different air pollutants and its effects on health and Environment.	10	2	2
5.	Briefly explain the following safety terminologies. i) Accident ii) Ergonomics iii) First aid iv) Hazard v) Safety inspection	10	2	2

Course outcomes:

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	26	34	-	-	10	50	-	-	-	-

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DEPARTMENT OF
BIOTECHNOLOGY
Academic year 2024-2025 (Odd Sem)

Date	22nd October 2024	Maximum Marks	60
Course Code	BT232AT	Duration	120 min
Sem	III Semester	Closed Book Offline Quiz & Test -1	

Bio Safety Standards and Ethics (Basket course)

Sl. No	Questions (Quiz)	M	BT-L	CO
1.	Covid 19 virus, will be categorized under which Bio safety level.	1	2	1
2.	Mention the composition of Institutional Biosafety Committee	1	2	2
3.	Define pathogen, give example.	1	1	1
4.	Mention the purpose of UV lamp in BSCs.	1	2	1
5.	List out the working mechanisms of HEPA filters.	2	3	1
6.	Expand the terms MERV and ULPA	2	3	1
7.	For the use of GMO at RVCE, what are the approvals required?	2	2	2

Sl. No	Questions (Test)	M	BT-L	CO
1.	Define Biohazard and mention the types of Biohazards with examples for each type.	10	2	1
2.	Explain biosafety levels with a neat triangular diagram, with examples for each of the Biosafety level.	10	4	1
3.	Illustrate on different types of Biosafety Cabinets, with the help of a neat diagram explain features and advantages of BSC- AII.	10	4	1
4.a 4.b	Define GMO. Explain with a neat diagram the production of protein rich tomato using carrot gene. Explain importance and applications of GMOs in various fields.	7 3	2	2
5.a 5.b	"Indian farmers, opposed the approval of Geneticaly Modified mustard seeds in the year 2022", mention the reasons for opposition and justify the statement in your own words. Define Cartagena protocol and mention its important functions.	7 3	4 2	2

BT-L-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	37	23	---	---	1	28	4	27	---	---

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DEPARTMENT OF CIVIL ENGINEERING

Date	8th Jan 2024	Maximum Marks	50
Course Code	CV232AT	Duration (minutes)	90
Sem	III Semester	CIE – I	
Environment and Sustainability			

o.	Questions	M	BT	CO
1	a) Define the terms i) Environment ii) Ecology iii) Bio-Diversity b) Materials flow in eco system is cyclic and Energy flow is unidirectional, justify.	6 4	1 2	1 1
2.	Classify different types of ecosystems. Explain in brief any two ecosystems with their structure and composition	10	2	2
3.	a) Recall the concept of ecological succession and classify the same b) Discuss the approaches of conserving the biodiversity	5 5	1 2	1 1
4.	a) Highlight the salient features of various hotspots of biodiversity in India b) Briefly discuss the causes, effects and preventive measures of water pollution	5 5	1 2	1 2
5.	Briefly describe the various Environmental protection acts in India	10	2	2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	25	25	--	-	16	34	-	-	-	-

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DEPARTMENT OF BIOTECHNOLOGY

Academic year 2024-2025 (Odd Sem)

Date	22nd October 2024	Maximum Marks	60
Course Code	BT232AT	Duration	120 min
Sem	III Semester	Closed Book Offline Quiz & Test -1	

Bio Safety Standards and Ethics (Basket course)

Sl. No	Questions (Quiz)	M	BT-L	CO
1.	Covid 19 virus, will be categorized under which Bio safety level.	1	2	1
2.	Mention the composition of Institutional Biosafety Committee	1	2	2
3.	Define pathogen, give example.	1	1	1
4.	Mention the purpose of UV lamp in BSCs.	1	2	1
5.	List out the working mechanisms of HEPA filters.	2	3	1
6.	Expand the terms MERV and ULPA	2	3	1
7.	For the use of GMO at RVCE, what are the approvals required?	2	2	2

Sl. No	Questions (Test)	M	BT-L	CO
1.	Define Biohazard and mention the types of Biohazards with examples for each type.	10	2	1
2.	Explain biosafety levels with a neat triangular diagram, with examples for each of the Biosafety level.	10	4	1
3.	Illustrate on different types of Biosafety Cabinets, with the help of a neat diagram explain features and advantages of BSC- AII.	10	4	1
4.a 4.b	Define GMO. Explain with a neat diagram the production of protein rich tomato using carrot gene. Explain importance and applications of GMOs in various fields.	7 3	2	2
5.a 5.b	"Indian farmers, opposed the approval of Geneticaly Modified mustard seeds in the year 2022", mention the reasons for opposition and justify the statement in your own words. Define Cartagena protocol and mention its important functions.	7 3	4 2	2

BT-L-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	37	23	---	---	1	28	4	27	---	---

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DEPARTMENT OF CIVIL ENGINEERING

Date	**8th Jan 2024**		
Course Code	**CV232AT**	Maximum Marks	**50**
Sem	**III Semester**	Duration (minutes)	**90**
	Environment and Sustainability	**CIE – I**	

No.	Questions	M	BT	CO
1	a) Define the terms i) Environment ii) Ecology iii) Bio-Diversity	6	1	1
	b) Materials flow in eco system is cyclic and Energy flow is unidirectional, justify.	4	2	1
2.	Classify different types of ecosystems. Explain in brief any two ecosystems with their structure and composition	10	2	2
3.	a) Recall the concept of ecological succession and classify the same	5	1	1
	b) Discuss the approaches of conserving the biodiversity	5	2	1
4.	a) Highlight the salient features of various hotspots of biodiversity in India	5	1	1
	b) Briefly discuss the causes, effects and preventive measures of water pollution	5	2	2
5.	Briefly describe the various Environmental protection acts in India	10	2	2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	25	25	--	-	16	34	-	-	-	-

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Tel: 080-68188185

Department of Civil Engineering

Date	21/10/2024	Maximum Marks	50
Course Code	CV232AT	Duration	90 Min +20 Min
Sem	III	CIE – I (Quiz + Test)	

Environment and Sustainability (Basket Course)

Part A

Sl. No.	Questions	M	BT	CO
1.	Distinguish between ecology and ecosystem.	2	1	1
2.	What is primary and secondary succession?	2	1	1
3.	Write any four methods of disposal of solid waste.	2	1	2
4.	What is the importance of safety in workplace?	2	1	2
5.	List any four environmental protection acts in India.	2	1	1

Part B

Sl. No.	Questions	M	BT	CO
1.	a. Briefly explain the structure and components of a ecosystem.	5	2	1
	b. Materials flow in eco system is bi directional and Energy flow is unidirectional justify.	5	2	1
2.	With a neat sketch explain the nitrogen cycle.	10	2	1
3.	Define solid waste management and list the objectives of solid waste management.	10	2	2
4.	Define Air Pollution? Classify and explain the different air pollutants and its effects on health and Environment.	10	2	2
5.	Briefly explain the following safety terminologies. i) Accident ii) Ergonomics iii) First aid iv) Hazard v) Safety inspection	10	2	2

Course outcomes:

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	26	34	-	-	10	50	-	-	-	-

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING & MANAGEMENT

Date	21st October 2024	Maximum Marks	10 + 50
Course Code	IM233AI	Duration	120 Mins
Sem	III Semester	CIE – I	

WORK SYSTEMS DESIGN

Note:
1. Answer all the Questions.

PART – A

Sl. No.	Questions	M	BT	CO
1	What is meant by the term Work?	02	2	1
2	Illustrate the Pyramidal Structure of the task.	02	3	1
3	Distinguish between methods design and methods engineering	02	2	2
4	Mention the elements of Standard Time.	02	2	1
5	What are the two common measures used in industry to assess a worker's productivity?	02	1	1

PART – B

Sl. No.	Questions	M	BT	CO
1.	Distinguish between Production and Productivity. Explain with suitable illustrations on the ways and means to enhance productivity in organizations.	10	3	CO1
2.	A work group of 10 workers in a certain month produced 1000 units of output working 8 hr/ day for 22 days in the month. What productivity measures could be used for this situation? Suppose that in the next month, the same work group produced 1200 units but there were only 20 work days in the month. Using the same productivity measures as before, determine the productivity index using the prior month as a base.	10	3	CO1
3.	The normal time to perform a repetitive manual task is 3.25 minutes. In addition, an irregular work element whose normal time is 1.5 minutes must be performed every 8 cycles. Two work units are produced each cycle. The PFD allowance is 12%. Determine the standard time per piece. How many work units are produced in an 8 hour shift at standard performance? Determine the anticipated amount of time worked and the amount of time lost per 8 hour shift that corresponds to the PFD allowance factor of 14%.	10	3	CO3
4.	Enumerate the steps involved in conducting a Method Study by an Industrial Engineer in organizations.	10	2	CO2
5 a.	Classify the worker –machine systems according to number of machines and workers.	05	1	CO1
b.	What does PFD stand for? Bring out the purpose of PFD allowance in determining the standard time for a task.	05	3	CO1

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max Marks	Quiz	08	02	--	--	--	02	06	02	--	--	--
		Test	30	10	10	--	--	05	10	35	--	--	--

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Academic year 2024 – 2025 (ODD Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING& MANAGEMENT

Date	23rd October 2024		
Course Code	IM235AI	Maximum Marks	10+50
Sem	III Semester	Duration	150 Min
		CIE – I	

DIGITAL METROLOGY

Part – A

Sl. No.	Questions	M	BT	CO
1.	The ______ stage may include amplification, filtering, and conversion of signals into a format that can be further processed.	01	L1	CO1
2.	The ___ in the intermediate stage transfers and amplifies the mechanical movement from the Bourdon tube.	01	L2	CO1
3.	For optical measuring systems, ______ can be an example for Detector Transducer Stage.	01	L2	CO1
4.	______ refers to the smallest change in a measured variable that an instrument can detect.	01	L1	CO1
5.	______ is the extent to which a measurement system can maintain its accuracy over time without recalibration.	01	L1	CO1
6.	______ is the minimum value of the input quantity to which an instrument responds and indicates a change in the output value.	01	L2	CO1
7.	The difference between the actual value of a changing quantity and the measured value due to the system's response characteristics is referred to as ______.	01	L2	CO1
8.	In the context of optical transducer, the primary function of a ______ is to sense light and convert it into a measurable electrical signal.	01	L2	CO2
9.	______ involves comparing the measurement output of an instrument directly against a known standard	01	L2	CO2
10.	A ______ is a type of resistive transducer that measures the deformation of an object by detecting changes in electrical resistance	01	L2	CO2

Part – B

Sl. No.	Questions	M	BT	CO
1.	Explain briefly the three stages of a generalized measurement system with block diagram. Consider any two examples and identify the three stages of measurement.	10	L2	CO1
2.	Describe briefly the important static and dynamic characteristics of Measurement systems with suitable sketches and graphs. How can the integration of both static and dynamic characteristics optimize the performance of a measurement system.	10	L2	CO1
3.	Describe the structure of an LVDT and explain how the movement of the core within the coils induces a voltage that is proportional to its position. Provide examples of various industrial applications where LVDTs are used for measurement.	10	L3	CO2
4.	Describe the working principle and applications of eddy current proximity sensors in metrology. Evaluate the effectiveness of eddy current proximity sensors in monitoring the rotational speed of turbine blades in high-speed industrial machinery, considering challenges and optimizations for optimal performance.	10	L3	CO2
5.	Explore the utilization of tachogenerators in metrology, focusing on their application in measuring rotational speed. Discuss the application of tachogenerators in providing real-time speed feedback within automated manufacturing systems.	10	L3	CO2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5
	Quiz	Max Marks	07	03	--	--	--	03	07	--	--	--
	Test		20	30	--	--	--	--	20	30	--	--

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF

INDUSTRIAL ENGINEERING & MANAGEMENT

Date	22nd October 2024	Maximum Marks	10 + 50
Course Code	IM234AI	Duration	120 Min
Sem	III Semester	CIE –I	
MANUFACTURING PROCESSES			

Note:
1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO
	Part – A			
1	__________ is the passage through which the molten metal from pouring basin reaches the mould cavity.	1	L1	CO1
2	________colour on surface of pattern is used to indicate the product has to be finished	1	L2	CO1
3	Balanced core is also called a __________ core.	1	L2	CO2
4	Loam sand is a mixture of	1	L1	CO2
5	______ Pattern is used for generating axi-symmerical and prismatic shapes.	1	L2	CO3
6	Coining is the operation of ________ forging	1	L1	CO2
7	List different types of moulding sand.	2	L1	CO3
8	Why chaplets are used in moulding process?	2	L2	CO2
	Part –B			
1	The foundry uses silica sand, mixed with bentonite clay as a binder, to create moulds for casting aluminum components. For a critical part, a complex valve body for an engine, the production has resulted in a high rejection rate due to defects. Your role is to conduct a case study based on a detailed analysis of the sand mould making procedure, assess the possible causes of these defect	10	L2	CO1
2	Explain the process used for pushing the billet or slug of metal through a die orifice, thus forming an elongated part of uniform cross section corresponding to the shape of the die?	10	L2	CO3
3	The foundry has been manufacturing engine components using sand casting. During quality checks, several castings were found to be either slightly undersized or oversized compared to the required dimensions. The investigation suggests that the allowances applied to the pattern might not be correctly calculated or uniformly applied. You have been tasked with reviewing the process and making recommendations to improve accuracy.	10	L3	CO2
4	"Jolt and squeeze moulding machine is used for preparing the moulds at faster rate and with high quality". Explain with a neat sketch.	10	L1	CO2
5	Explain the working principle of Hydro-forming Process with its limitations.	10	L2	CO1

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	03	05	02	10	04	06	--	--	--	--
	Test		20	20	10	--	10	30	10	--	--	--

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Academic year 2024 – 2025 (ODD Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING& MANAGEMENT

Date	23rd October 2024		
Course Code	IM235AI	Maximum Marks	10+50
Sem	III Semester	Duration	150 Min
		CIE – I	

DIGITAL METROLOGY

Part – A

Sl. No.	Questions	M	BT	CO
1.	The ______ stage may include amplification, filtering, and conversion of signals into a format that can be further processed.	01	L1	CO1
2.	The ____ in the intermediate stage transfers and amplifies the mechanical movement from the Bourdon tube.	01	L2	CO1
3.	For optical measuring systems, ______ can be an example for Detector Transducer Stage.	01	L2	CO1
4.	______ refers to the smallest change in a measured variable that an instrument can detect.	01	L1	CO1
5.	______ is the extent to which a measurement system can maintain its accuracy over time without recalibration.	01	L1	CO1
6.	______is the minimum value of the input quantity to which an instrument responds and indicates a change in the output value.	01	L2	CO1
7.	The difference between the actual value of a changing quantity and the measured value due to the system's response characteristics is referred to as ______.	01	L2	CO1
8.	In the context of optical transducer, the primary function of a ______ is to sense light and convert it into a measurable electrical signal.	01	L2	CO2
9.	______ involves comparing the measurement output of an instrument directly against a known standard	01	L2	CO2
10.	A ________ is a type of resistive transducer that measures the deformation of an object by detecting changes in electrical resistance	01	L2	CO2

Part – B

Sl. No.	Questions	M	BT	CO
1.	Explain briefly the three stages of a generalized measurement system with block diagram. Consider any two examples and identify the three stages of measurement.	10	L2	CO1
2.	Describe briefly the important static and dynamic characteristics of Measurement systems with suitable sketches and graphs. How can the integration of both static and dynamic characteristics optimize the performance of a measurement system.	10	L2	CO1
3.	Describe the structure of an LVDT and explain how the movement of the core within the coils induces a voltage that is proportional to its position. Provide examples of various industrial applications where LVDTs are used for measurement.	10	L3	CO2
4.	Describe the working principle and applications of eddy current proximity sensors in metrology. Evaluate the effectiveness of eddy current proximity sensors in monitoring the rotational speed of turbine blades in high-speed industrial machinery, considering challenges and optimizations for optimal performance.	10	L3	CO2
5.	Explore the utilization of tachogenerators in metrology, focusing on their application in measuring rotational speed. Discuss the application of tachogenerators in providing real-time speed feedback within automated manufacturing systems.	10	L3	CO2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5
	Quiz	Max Marks	07	03	--	--	--	03	07	--	--	--
	Test		20	30	--	--	--	--	20	30	--	--

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Academic year 2023-2024 (Odd Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	9th Jan 2024	Maximum Marks	50
Course Code	IM234AI	Duration	90 Min
Sem	III Semester	CIE – 1	
MANUFACTURING PROCESS			

Note:
1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO
1.	Discuss the objectives expected to be achieved through efficient and economical machining?	10	1	1
2.	Draw a neat sketch of a single point cutting tool indicating its complete geometry.	10	2	2
3.	How many types of chips are formed in metal cutting? List the factors that are responsible for formation of these chips.	10	3	3
4.	Explain crater wear and flank wear of a tool with a neat sketch.	10	2	2
5.	In orthogonal cutting of a work piece, the following data were obtained: Feed force = 350 kgf. Cutting force = 150 kgf Chip thickness ratio = 0.44 Rake angle = 10° Calculate the shear angle, shear force and normal force.	10	3	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	10	20	20	--	10	20	20	--	--	--

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Academic year 2021-2002 (Odd Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING & MANAGEMENT

Date	9th Jan 2024		
Course Code	IM233AI	Maximum Marks	50
Sem	III Semester	Duration	90 Min
		CIE – I	

WORK SYSTEMS DESIGN

Note:
1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO
1.	Analyze the three categories of excess non-productive activities generally carried out in organizations.	10	4	CO1
2.	A work group of 10 workers in a certain month produced 7200 units of output working 8 hours per day for 22 days in the month. Determine the labor productivity ratio using units of output per worker hour and units of output per worker month. Suppose that in the next month, the same work group produced 6800 units but there were only 20 workdays in the month .For each productivity measure, determine the productivity index for the next month using prior month as a base.	10	3	CO1
3.	The normal time to perform a certain manual work cycle is 3.47 minutes. In addition, an irregular work element whose normal time is 3.70 minutes must be performed every 10 cycles. One work unit is produced each cycle. The PFD allowance factor is 14%.Determine the standard time per piece and how many work units are produced during an 8 hour shift at 100 percent performance and the worker works exactly 7.018 hours, which corresponds to 14 percent allowance factor. If the workers performance is 120 percent and works 7.2 hours during regular shift, how many units are produced?	10	3	CO3
4.	Distinguish between Production and Productivity. Also justify the rationale for considering labor productivity as a measure of organizational efficiency and effectiveness.	10	3	CO1
5 a.	Enumerate any five reasons for variations in work cycle times in manual work systems.	05	2	CO1
b.	Illustrate the classification of powered machinery in worker machine system.	05	3	CO1

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Test	Max Marks	40	---	10	--	---	--	05	35	10	--	--

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Academic year 2023-24 (ODD Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING& MANAGEMENT

Date	10th January 2024	**Maximum Marks**	50
Course Code	IM235AI	**Duration**	90 Min
Sem	III Semester - 2022 scheme	CIE – I	

DIGITAL METROLOGY

Note:
1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO
1.	Explain briefly the three stages of a generalized measurement system with block diagram. Consider any two examples and identify the three stages of measurement.	(10)	L2	CO1
2.	Describe briefly the important static and dynamic characteristics of Measurement systems with suitable sketches and graphs. How can the integration of both static and dynamic characteristics optimize the performance of a measurement system, and what considerations should be taken into account to ensure a balanced and effective design	(10)	L3	CO1
3.	Enumerate briefly on different types of Standards and Calibration. Explain the role of calibration in ensuring accuracy and reliability in measurements, and how do standards contribute to the traceability of measurements in various industries.	(10)	L3	CO1
4.	Explain the principle of a Resistive Transducers. Explain any two Resistive Transducers with neat sketches.	(10)	L2	CO1
5.	Discuss briefly on the statistical Evaluation of measured data. What role does statistical analysis play in assessing the accuracy and precision of measured data, provide examples of specific statistical tests commonly used for this purpose.	(10)	L2	CO1

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Test	Max Marks	50	--	--	--	--	--	30	20	--	--	--

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Academic year 2021-2002 (Odd Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING & MANAGEMENT

Date	9th Jan 2024		
Course Code	IM233AI	Maximum Marks	50
Sem	III Semester	Duration	90 Min
		CIE – I	
WORK SYSTEMS DESIGN			

Note:

1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO
1.	Analyze the three categories of excess non-productive activities generally carried out in organizations.	10	4	CO1
2.	A work group of 10 workers in a certain month produced 7200 units of output working 8 hours per day for 22 days in the month. Determine the labor productivity ratio using units of output per worker hour and units of output per worker month. Suppose that in the next month, the same work group produced 6800 units but there were only 20 workdays in the month .For each productivity measure, determine the productivity index for the next month using prior month as a base.	10	3	CO1
3.	The normal time to perform a certain manual work cycle is 3.47 minutes. In addition, an irregular work element whose normal time is 3.70 minutes must be performed every 10 cycles. One work unit is produced each cycle. The PFD allowance factor is 14%.Determine the standard time per piece and how many work units are produced during an 8 hour shift at 100 percent performance and the worker works exactly 7.018 hours, which corresponds to 14 percent allowance factor. If the workers performance is 120 percent and works 7.2 hours during regular shift, how many units are produced?	10	3	CO3
4.	Distinguish between Production and Productivity. Also justify the rationale for considering labor productivity as a measure of organizational efficiency and effectiveness.	10	3	CO1
5 a.	Enumerate any five reasons for variations in work cycle times in manual work systems.	05	2	CO1
b.	Illustrate the classification of powered machinery in worker machine system.	05	3	CO1

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Test	Max Marks	40	---	10	--	--	--	05	35	10	--	--

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Academic year 2023-2024 (Odd Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	9^{th} Jan 2024	Maximum Marks	50
Course Code	IM234AI	Duration	90 Min
Sem	III Semester	CIE – 1	
MANUFACTURING PROCESS			

Note:
1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO
1.	Discuss the objectives expected to be achieved through efficient and economical machining?	10	1	1
2.	Draw a neat sketch of a single point cutting tool indicating its complete geometry.	10	2	2
3.	How many types of chips are formed in metal cutting? List the factors that are responsible for formation of these chips.	10	3	3
4.	Explain crater wear and flank wear of a tool with a neat sketch.	10	2	2
5.	In orthogonal cutting of a work piece, the following data were obtained: Feed force = 350 kgf. Cutting force = 150 kgf Chip thickness ratio = 0.44 Rake angle = 10° Calculate the shear angle, shear force and normal force.	10	3	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	10	20	20	--	10	20	20	--	--	--

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	22nd October 2024	Maximum Marks	10 + 50
Course Code	IM234AI	Duration	120 Min
Sem	III Semester	CIE –I	
MANUFACTURING PROCESSES			

Note:
1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO
	Part – A			
1	__________ is the passage through which the molten metal from pouring basin reaches the mould cavity.	1	L1	CO1
2	________colour on surface of pattern is used to indicate the product has to be finished	1	L2	CO1
3	Balanced core is also called a ____________ core.	1	L2	CO2
4	Loam sand is a mixture of ________	1	L1	CO2
5	_____ Pattern is used for generating axi-symmerical and prismatic shapes.	1	L2	CO3
6	Coining is the operation of ________ forging	1	L1	CO2
7	List different types of moulding sand.	2	L1	CO3
8	Why chaplets are used in moulding process?	2	L2	CO2
	Part –B			
1	The foundry uses silica sand, mixed with bentonite clay as a binder, to create moulds for casting aluminum components. For a critical part, a complex valve body for an engine, the production has resulted in a high rejection rate due to defects. Your role is to conduct a case study based on a detailed analysis of the sand mould making procedure, assess the possible causes of these defect	10	L2	CO1
2	Explain the process used for pushing the billet or slug of metal through a die orifice, thus forming an elongated part of uniform cross section corresponding to the shape of the die?	10	L2	CO3
3	The foundry has been manufacturing engine components using sand casting. During quality checks, several castings were found to be either slightly undersized or oversized compared to the required dimensions. The investigation suggests that the allowances applied to the pattern might not be correctly calculated or uniformly applied. You have been tasked with reviewing the process and making recommendations to improve accuracy.	10	L3	CO2
4	"Jolt and squeeze moulding machine is used for preparing the moulds at faster rate and with high quality". Explain with a neat sketch.	10	L1	CO2
5	Explain the working principle of Hydro-forming Process with its limitations.	10	L2	CO1

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	03	05	02	10	04	06	--	--	--	--
	Test		20	20	10	--	10	30	10	--	--	--

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Academic year 2022-2023 (ODD Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING & MANAGEMENT

Date	18th January 2023	Maximum Marks	10 + 50
Course Code	21IM33	Duration	20 + 90 Min
Sem	III Semester	CIE – I	

WORK SYSTEMS DESIGN

Note:
1. Answer all the Questions.

PART – A

Sl. No.	Questions	M	BT	CO
1	What is meant by the term Work?	02	2	1
2	Illustrate the Pyramidal Structure of the task.	02	3	1
3	How is Motion Study different from methods engineering?	02	2	2
4	What is meant by productivity?	02	2	1
5	List the three categories of principles of motion economy.	02	1	2

PART – B

Sl. No.	Questions	M	BT	CO
1.	Analyze the problems encountered by organizations in measuring productivity.	10	4	CO1
2.	A work group of 5 workers in a certain month produced 500 units of output working 8 hr/ day for 22 days in the month. What productivity measures could be used for this situation? Suppose that in the next month, the same work group produced 600 units but there were only 20 work days in the month. Using the same productivity measures as before, determine the productivity index using the prior month as a base.	10	3	CO1
3.	Enumerate the principles of motion economy related to the use of Human body.	10	2	CO2
4.	Methods engineering can be categorized into Methods analysis and Methods design. Distinguish between them in terms of their scope, objectives and applications.	10	3	CO2
5.	The normal time to perform a repetitive manual task is 4.25 minutes. In addition, an irregular work element whose normal time is 1.75 minutes must be performed every 8 cycles. Two work units are produced each cycle. The PFD allowance is 16%. Determine the standard time per piece. How many work units are produced in an 8 hour shift at standard performance? Determine the anticipated amount of time worked and the amount of time lost per 8 hour shift that corresponds to the PFD allowance factor of 16%.	10	3	CO3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Test	Max Marks	06	04	--	--	--	02	06	02	--	--	--
	Quiz		20	20	10	--	--	--	10	30	10	--	--

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Academic year 2022-2023 (ODD Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING & MANAGEMENT

Date	18th January 2023	Maximum Marks	10 + 50
Course Code	21IM33	Duration	20 + 90 Min
Sem	III Semester	CIE – I	

WORK SYSTEMS DESIGN

Note:

1. Answer all the Questions.

PART – A

Sl. No.	Questions	M	BT	CO
1	What is meant by the term Work?	02	2	1
2	Illustrate the Pyramidal Structure of the task.	02	3	1
3	How is Motion Study different from methods engineering?	02	2	2
4	What is meant by productivity?	02	2	1
5	List the three categories of principles of motion economy.	02	1	2

PART – B

Sl. No.	Questions	M	BT	CO
1.	Analyze the problems encountered by organizations in measuring productivity.	10	4	CO1
2.	A work group of 5 workers in a certain month produced 500 units of output working 8 hr/ day for 22 days in the month. What productivity measures could be used for this situation? Suppose that in the next month, the same work group produced 600 units but there were only 20 work days in the month. Using the same productivity measures as before, determine the productivity index using the prior month as a base.	10	3	CO1
3.	Enumerate the principles of motion economy related to the use of Human body.	10	2	CO2
4.	Methods engineering can be categorized into Methods analysis and Methods design. Distinguish between them in terms of their scope, objectives and applications.	10	3	CO2
5.	The normal time to perform a repetitive manual task is 4.25 minutes. In addition, an irregular work element whose normal time is 1.75 minutes must be performed every 8 cycles. Two work units are produced each cycle. The PFD allowance is 16%. Determine the standard time per piece. How many work units are produced in an 8 hour shift at standard performance? Determine the anticipated amount of time worked and the amount of time lost per 8 hour shift that corresponds to the PFD allowance factor of 16%.	10	3	CO3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Test	Max Marks	06	04	--	--	--	02	06	02	--	--	--
	Quiz		20	20	10	--	--	--	10	30	10	--	--

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Academic year 2023-24 (ODD Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING& MANAGEMENT

Date	10th January 2024		
Course Code	IM235AI	Maximum Marks	50
Sem	III Semester - 2022 scheme	Duration	90 Min
		CIE – I	

DIGITAL METROLOGY

Note:

1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO
1.	Explain briefly the three stages of a generalized measurement system with block diagram. Consider any two examples and identify the three stages of measurement.	(10)	L2	CO1
2.	Describe briefly the important static and dynamic characteristics of Measurement systems with suitable sketches and graphs. How can the integration of both static and dynamic characteristics optimize the performance of a measurement system, and what considerations should be taken into account to ensure a balanced and effective design	(10)	L3	CO1
3.	Enumerate briefly on different types of Standards and Calibration. Explain the role of calibration in ensuring accuracy and reliability in measurements, and how do standards contribute to the traceability of measurements in various industries.	(10)	L3	CO1
4.	Explain the principle of a Resistive Transducers. Explain any two Resistive Transducers with neat sketches.	(10)	L2	CO1
5.	Discuss briefly on the statistical Evaluation of measured data. What role does statistical analysis play in assessing the accuracy and precision of measured data, provide examples of specific statistical tests commonly used for this purpose.	(10)	L2	CO1

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Test	Max Marks	50	--	--	--	--	--	30	20	--	--	--

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING & MANAGEMENT

Date	21st October 2024	Maximum Marks	10 + 50
Course Code	IM233AI	Duration	120 Mins
Sem	III Semester	CIE – I	
WORK SYSTEMS DESIGN			

Note:
1. Answer all the Questions.

PART – A

Sl. No.	Questions	M	BT	CO
1	What is meant by the term Work?	02	2	1
2	Illustrate the Pyramidal Structure of the task.	02	3	1
3	Distinguish between methods design and methods engineering	02	2	2
4	Mention the elements of Standard Time.	02	2	1
5	What are the two common measures used in industry to assess a worker's productivity?	02	1	1

PART – B

Sl. No.	Questions	M	BT	CO
1.	Distinguish between Production and Productivity. Explain with suitable illustrations on the ways and means to enhance productivity in organizations.	10	3	CO1
2.	A work group of 10 workers in a certain month produced 1000 units of output working 8 hr/ day for 22 days in the month. What productivity measures could be used for this situation? Suppose that in the next month, the same work group produced 1200 units but there were only 20 work days in the month. Using the same productivity measures as before, determine the productivity index using the prior month as a base.	10	3	CO1
3.	The normal time to perform a repetitive manual task is 3.25 minutes. In addition, an irregular work element whose normal time is 1.5 minutes must be performed every 8 cycles. Two work units are produced each cycle. The PFD allowance is 12%. Determine the standard time per piece. How many work units are produced in an 8 hour shift at standard performance? Determine the anticipated amount of time worked and the amount of time lost per 8 hour shift that corresponds to the PFD allowance factor of 14%.	10	3	CO3
4.	Enumerate the steps involved in conducting a Method Study by an Industrial Engineer in organizations.	10	2	CO2
5 a.	Classify the worker –machine systems according to number of machines and workers.	05	1	CO1
b.	What does PFD stand for? Bring out the purpose of PFD allowance in determining the standard time for a task.	05	3	CO1

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max Marks	Quiz	08	02	--	--	--	02	06	02	--	--	--
		Test	30	10	10	--	--	05	10	35	--	--	--

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DEPARTMENT OF MATHEMATICS

Academic year 2024-2025 (Odd Semester 2024)

Date	21.10.2024	Time	9.30 AM– 11.30 AM	
TEST	Quiz- I & Test -I	Maximum Marks	10+50	
Course Title	STATISTICS, LAPLACE TRANSFORM AND NUMERICAL METHODS		Course Code	MA231TB
Semester	III	Programs	AS, BT, CH, ME, IEM	

Sl. No	Instructions: Answer all questions. Part - A	M	CO	BT
1	For a statistical data with $n = 5, \sum x = 18, \sum y = 15, \sum x^2 = 110$ and $\sum xy = 71$, the best fit straight line by the method of least squares is ________.	1	1	1
2	For a distribution the mean is 12 and the variance is 19. Then the second moment about the mean is	1	2	1
3	The first two moments about the point $x = 2$ are 1 and 16, then the second central moment is ________.	2	2	2
4	The equations of regression lines are $y = 0.5x + p$ and $y = 0.4y + q$, then the correlation coefficient is ________.	2	2	2
5	$L[\sin 3t + \cosh 5t] =$ ________.	1	1	1
6	$L[(t^2 e^{3t})] =$ ________.	1	2	1
7	$L[(e^{-5t} + e^{t})^2] =$ ________.	2	1	2
	Part - B			
1	Determine the equation of the regression plane connecting x_1, x_2 and y, given (table below). Estimate y at $x_1 = 1.8$ and $x_2 = 112$.	10	4	3
2	The marks obtained by 50 students in a Mathematics test are given below (table below). Compute first four moments about the mean and also find the measures β_1 and β_2 for the following distribution and comment on the nature of distribution of students.	10	3	
3a	The following pairs of observations were noted in an experimental work on cosmic rays. Find by the method of least squares the most possible values of a and b for the equation $y = a + bx$ which fits the data (table below). Predict y when $x = 3.75$.	6	1	

Part B, Question 1:

Diffusion time (hours) x_1	1.5	2.5	0.5	1.2	2.6	0.3
Sheet – resistance (Ohms – cm) x_2	66	87	69	141	93	105
Current gain (y)	5.3	7.8	7.4	9.8	10.8	9.1

Estimate y at $x_1 = 1.8$ and $x_2 = 112$.

Part B, Question 2:

Marks	10-20	20-30	30-40	40-50	50-60	60-70
No. of students	4	8	10	15	8	5

Part B, Question 3a:

x	1	2	3	4	5	6
y	2.98	4.26	5.21	6.10	6.80	7.50

Predict y when $x = 3.75$.

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Academic year 2022-2023 (Odd Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING AND MANAGEMENT

Date	**19th January 2023**	**Maximum Marks**	**10+50**
Course Code	**21IM34**	**Duration**	**20+90 Min**
Sem	**III Semester**	**CIE - I**	

MANUFACTURING PROCESS

Note:
1. Answer all the Questions.

PART – A

QNo.1	Questions	M	BT	CO
1.1	In ________ forging operation, repeated hammering and closed die is used.	01	L2	CO1
1.2	________ Molding machine gives better packing of sand near pattern	01	L1	CO2
1.3	In sand molding, the bottom most part of the flask is called________	01	L2	CO1
1.4	Over hang core is also called as ________	01	L1	CO2
1.5	________ Pattern are generally used for circular castings	01	L2	CO1
1.6	Irregular depressions on the surface of the forging is the ____ defect in forging	01	L3	CO2
1.7	Define porosity?	02	L1	CO1
1.8	List any two limitations of Extrusion process	02	L2	CO2

PART – B

Sl. No.	Questions	M	BT	CO
1.	Define pattern? List the types of patterns and explain any two briefly.	10	L2	CO2
2.	"Sand slinger moulding machine is used for preparing the moulds at faster rate and with high quality". Explain with a neat sketch.	10	L2	CO1
3.	"Molding processes to produce internal cavities and reentrant angles". What is the process called and explain the different types.	10	L3	CO2
4.	Explain the stages in drop forging a connecting rod for an internal combustion engine	10	L3	CO1
5.	Briefly explain the principle of Rolling with a neat sketch	10	L3	CO1

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
Marks Distribution	Quiz	Max Marks	05	05	--	--	04	05	01	--	--	--
	Test		30	20	--	--	10	20	20	--	--	--

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Department of Industrial Engineering and Management

CONTINUOUS INTERNAL EVALUATION I – September 2019

III SEMESTER

METROLOGY AND MEASUREMENTS - 18IM33

DURATION: 90 min **MAX. MARKS: 50**

Course Outcomes:

CO1:	Discuss the principles and practices of metrology in manufacturing environment and analyze uncertainty in an appropriate manner.
CO2:	Describe the operating principles of range of widely used instrumentation techniques and illustrate how to use them in the design of measurement systems.
CO3:	Compare the production process, the product function and the product design, and to select appropriate measurement quantities and tools for these purposes.
CO4:	Evaluate and respond to the need for rigorous and formal metrology concepts in designing and using measurement systems

Q. No	Questions	M	CO	BT
Q1.	With an example, enumerate on the generalized measurement system. Justify the answer by considering bourdon tube as an example. Describe with neat sketches and block diagrams.	(10)	L2	CO1
Q2a	Describe briefly the different static characteristics of Measurement systems.	(06)	L2	CO2
b	Define Error. Explain briefly the different types of Errors and remedies to resolve them.	(04)	L2	CO2
Q3a	Define (i) Maximum Clearance (ii) Basic Shaft (iii) Fundamental Deviation and schematically represent it.	(06)	L1	CO1
b	Enumerate on the concept (i) Interchangeability (ii) Hole basis systems	(04)	L2	CO2
Q4.	Determine the tolerances on the hole and the shaft for a precision running fit designated by 25H7g6, given; • 25 mm lies between 18-30 mm • Fundamental deviation for g shaft $=-16D^{0.44}$ • IT7=16i and IT6=10i State the actual maximum and minimum sizes of the hole and shaft and maximum and minimum clearances and represent all the dimensions graphically.	(10)	L4	CO2
Q5a	Define Fit. Explain briefly the various types of fits with examples.	(10)	L2	CO2

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Academic year 2022-2023 (Odd Sem)

DEPARTMENT OF

INDUSTRIAL ENGINEERING AND MANAGEMENT

Date	19th January 2023	Maximum Marks	10+50
Course Code	21IM34	Duration	20+90 Min
Sem	III Semester	CIE - I	

MANUFACTURING PROCESS

Note:

1. Answer all the Questions.

PART – A

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1.2	__________ Molding machine gives better packing of sand near pattern	01	L1	CO2
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1.7	Define porosity?	02	L1	CO1
1.8	List any two limitations of Extrusion process	02	L2	CO2

PART – B

Sl. No.	Questions	M	BT	CO
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BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz		05	05	--	--	04	05	01	--	--	--
	Test	Max Marks	30	20	--	--	10	20	20	--	--	--

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Department of Industrial Engineering and Management
CONTINUOUS INTERNAL EVALUATION I – September 2019
III SEMESTER
METROLOGY AND MEASUREMENTS - 18IM33

DURATION: 90 min **MAX. MARKS: 50**

Course Outcomes:

CO1:	Discuss the principles and practices of metrology in manufacturing environment and analyze uncertainty in an appropriate manner.
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CONTINUOUS INTERNAL EVALUATION - I – Sept. 2019

III SEMESTER

18IM35 – WORK SYSTEMS DESIGN

DURATION: 90 min MAX. MARKS:50

Instructions to Students:

1. Answer all questions from Part B.

Course Outcomes:	
CO1	State the industrial engineering principles that influence the productivity improvement in organizations.
CO2	Apply the methods engineering and operational analysis in re-designing of work systems.
CO3	Apply engineering work measurement principles in analyzing and measurement of work.
CO4	Analyze the work processes using advanced work study tools and techniques.
CO5	Demonstrate an understanding of emerging concepts and applications in designing work systems.

Part – B **Max. Marks 50**

Q.No	Questions	M	BT	CO
Q1 a	Distinguish between Labour productivity ratio and Labour Productivity Index.	(05)	L2	CO1
b	Distinguish between work system as a physical entity and work system as a profession.	(05)	L2	CO1
Q2	A work group of 5 workers in a certain month produced 500 units of output working 8 hr/day for 22 days in the month. (i) What productivity measures could be used for this situation, and what are the values of their respective productivity ratios? (ii) Suppose that in the next month, the same work group produced 600 units but there were only 20 work days in the month. Using the same productivity measures as before, determine the productivity index using the prior month as a base.	(10)	L3	CO1
Q3	The normal time to perform a certain manual work cycle is 3.47 min. In addition, an irregular work element whose normal time is 3.70 min must be performed every 10 cycles. One work unit is produced each cycle. The PFD allowance factor is 14%. Determine (i) the standard time per piece and (b) how many work units are produced during an 8-hour shift at 100% performance, and the worker works exactly 7.018 hr, which corresponds to the 14% allowance factor. (iii) If the worker's pace is 120% and he works 7.2 hours during the regular shift, how many units are produced?	(10)	L4	CO3
Q4	Distinguish between the following along with an example for each. (i) Office automation vs. Office augmentation (ii) Office activities vs. Office applications	 (05) (05)	L2	CO1
Q5	Review any five principles related to the design of tooling and equipment.	(10)	L4	CO2

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M - Marks, BT - Blooms Taxanomy level, CO - Course outcome

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CONTINUOUS INTERNAL EVALUATION - I – Sept. 2019

III SEMESTER

18IM35 – WORK SYSTEMS DESIGN

DURATION: 90 min | **MAX. MARKS:50**

Instructions to Students:

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Course Outcomes:	
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M - Marks, BT - Blooms Taxanomy level, CO - Course outcome

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CONTINUOUS INTERNAL EVALUATION- I –September 2019

III SEMESTER

18IM36 – Manufacturing Process

DURATION: 90 min | MAX. MARKS: 50

Instructions to Students:

1. Answer all questions from Part B.

Course outcomes:

CO1	Explain the basic principles and methodology of various manufacturing processes that are used for the production of mechanical parts and products.
CO2	Compare and contrast the advantages and limitations of different manufacturing processes.
CO3	Solve the problems on processing time and economics of processing of material with respect to a manufacturing process.
CO4	Apply the design concept of various manufacturing processes when a specific product has to be manufactured.

Part – B **Max. Marks 50**

Q.No	Questions	M	BT	CO
Q1	Discuss orthogonal & oblique cutting with respect to the following: i. Angle of contact ii. Depth of cut iii. Chip form iv. Area of contact	(10)	2	2
Q2	State the assumptions made by Merchant while constructing a Merchant Circle diagram.	(10)	2	2
Q3	In an orthogonal turning operations on a lathe the following data were obtained: Cutting force = 120 kg, Feed force = 30 kg, Back rake angle = 15°, Feed rate = 0.2 mm/rev, chip thickness = 0.3 mm, Cutting speed = 100 m/min, Work piece diameter = 120 mm, depth of cut = 0.4 mm. Calculate chip thickness ratio, shear angle, coefficient of friction, friction angle, shear stress, shear strain, strain energy and chip flow velocity.	(10)	1	1
Q4	In a tool wear test with a high speed steel cutting tool, the following values of tool life were obtained: Tool Life: 30 min – Cutting Speed: 25.0 m/min Tool Life: 1.5 min – Cutting Speed: 70.0 m/min Calculate the values of 'n' & 'c' using Taylor's Tool Life equation.	(10)	1	1
Q5	Discuss the factors which affect the parameters of Tool Life.	(10)	2	2

Tool Life	Cutting Speed
30 min	25.0 m/min
1.5 min	70.0 m/min

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Note: M - Marks, BT - Blooms Taxonomy Level, CO - Course Outcome

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CONTINUOUS INTERNAL EVALUATION- I –September 2019

III SEMESTER

18IM36 – Manufacturing Process

DURATION: 90 min **MAX. MARKS: 50**

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1. Answer all questions from Part B.

Course outcomes:

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Note: M - Marks, BT - Blooms Taxonomy Level, CO - Course Outcome